

Discrete Random Variables

Department of Computer Science

University of Botswana

August 21, 2026



Learning Objectives

By the end of this lecture, you should be able to:

- define a random variable and distinguish discrete from continuous random variables;
- construct a PMF from a sample space;
- check whether a function is a valid PMF;
- compute probabilities, expectation, variance and standard deviation;
- recognize common discrete distributions: Bernoulli, uniform, binomial, geometric and Poisson;
- explain joint PMFs, marginal PMFs, conditional PMFs and independence using tables;
- solve exam-style problems involving discrete random variables.

Motivation

Many experiments have outcomes that are not numbers.

Examples

- Tossing coins: outcome can be HHT, TTH, etc.
- Rolling two dice: outcome can be $(3, 5)$.
- Observing a system: outcome can be success or failure.

Key idea

A random variable assigns a number to each outcome so that we can calculate probabilities using numbers.

Random Variables: The Idea

Definition

A **random variable** is a function that assigns a real number to every outcome in the sample space.

$$X : \Omega \longrightarrow \mathbb{R}$$

Example: two dice

Let the outcome be (d_1, d_2).

- $X = d_1 + d_2$ is the sum of the two dice.
- $Y = |d_1 - d_2|$ is the absolute difference.
- $Z = \max(d_1, d_2)$ is the maximum.

Notation

- Use capital letters for random variables: X, Y, Z.
- Use lowercase letters for values: x, y, z.
- The statement $X = x$ is an event.

Example

If X is the sum of two dice, then

$$\{X = 7\} = \{(1,6), (2,5), (3,4), (4,3), (5,2), (6,1)\}.$$

Therefore,

$$\mathbb{P}(X = 7) = \frac{6}{36} = \frac{1}{6}.$$

Discrete and Continuous Random Variables

Discrete random variable

A random variable is **discrete** if it takes values in a finite or countably infinite set.

Continuous random variable

A random variable is **continuous** if it can take values in an interval of real numbers.

Classify

- Number of heads in 3 coin tosses: discrete.
- Sum of two dice: discrete.
- Time until the next request arrives at a server: continuous.
- CPU processing time for a request: continuous.

From Random Variables to Events

When we set a random variable equal to a value, we get an event.

Random variable	Event	Probability
$X =$ sum of two dice	$X = 7$	$\mathbb{P}(X = 7) = 1/6$
$N =$ arrivals in one hour	$N > 10$	$\mathbb{P}(N > 10)$
$Y =$ heads in 3 tosses	$Y = 2$	$\mathbb{P}(Y = 2) = 3/8$

Important

The probability rules for events still apply to random variables.

Example: Three Fair Coins

We flip three fair coins and define

$$Y = \text{total number of heads observed.}$$

Possible values:

$$Y \in \{0, 1, 2, 3\}.$$

Note

$Y = 4$ is impossible because only three coins are tossed.

Sample Space for Three Coin Tosses

The sample space has $2^3 = 8$ equally likely outcomes:

$$\Omega = \{HHH, HHT, HTH, THH, HTT, THT, TTH, TTT\}.$$

Each outcome has probability

$$\frac{1}{8}.$$

Counting idea

To find $\mathbb{P}(Y = k)$, count the outcomes with exactly k heads.

PMF of Y: Number of Heads in Three Tosses

y	Outcomes	$p_Y(y) = \mathbb{P}(Y = y)$
0	TTT	$1/8$
1	HTT, THT, TTH	$3/8$
2	HHT, HTH, THH	$3/8$
3	HHH	$1/8$

$$\mathbb{P}(Y = k) = \frac{\binom{3}{k}}{8}, \quad k = 0, 1, 2, 3.$$

Probability Mass Function

Definition

The **probability mass function** (PMF) of a discrete random variable X assigns probabilities to possible values of X.

$$p_X(x) = \mathbb{P}(X = x)$$

PMF properties

A function $p_X(x)$ is a valid PMF if

$$p_X(x) \geq 0 \quad \text{for all } x$$

and

$$\sum_x p_X(x) = 1.$$

Checking Whether a PMF is Valid

Suppose

$$p_X(0) = 0.2, \quad p_X(1) = 0.5, \quad p_X(2) = 0.3.$$

Check 1: non-negative probabilities

All probabilities are non-negative.

Check 2: probabilities sum to 1

$$0.2 + 0.5 + 0.3 = 1.$$

Conclusion

This is a valid PMF.

Example: Rolling One Fair Die

Let $X =$ outcome of rolling a fair six-sided die.

$$X \in \{1, 2, 3, 4, 5, 6\}.$$

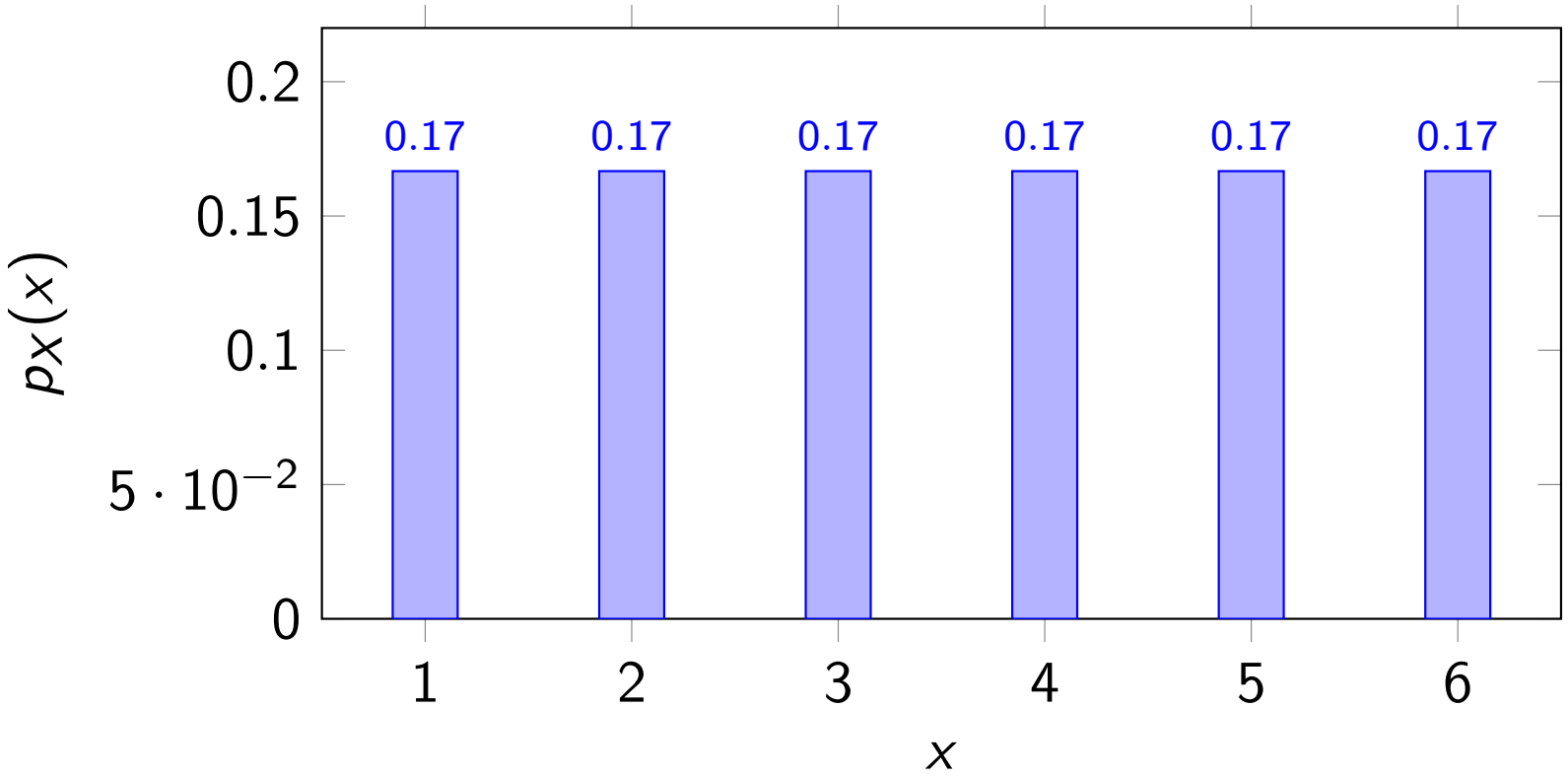
Since all outcomes are equally likely,

$$p_X(x) = \mathbb{P}(X = x) = \frac{1}{6}, \quad x = 1, 2, 3, 4, 5, 6.$$

Otherwise,

$$p_X(x) = 0.$$

PMF Bar Chart: Fair Die



Example: Sum of Two Fair Dice

Let

$$X = d_1 + d_2.$$

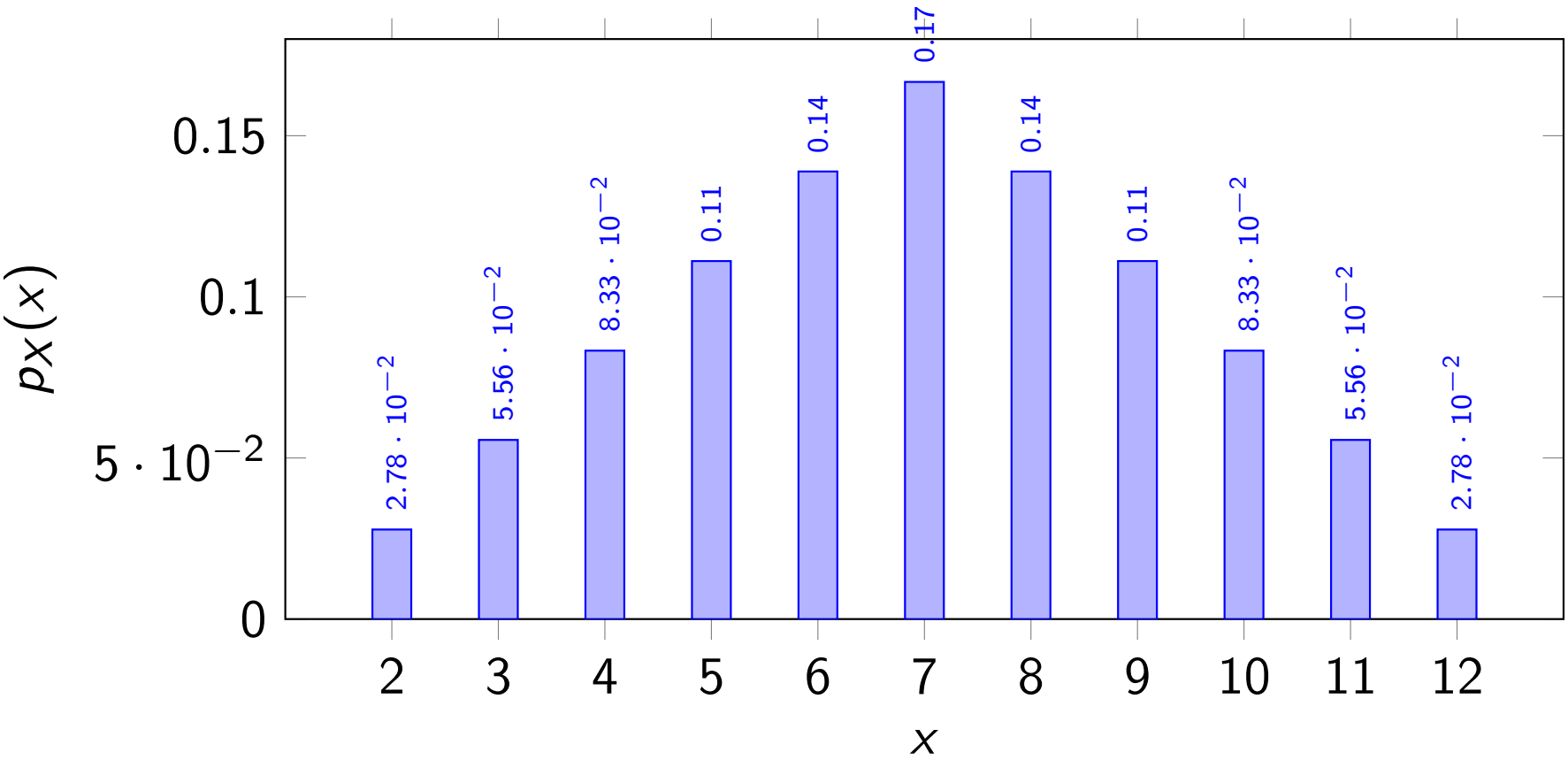
There are 36 equally likely outcomes.

Counting formula

$$p_X(x) = \frac{\#\{(d_1, d_2) : d_1 + d_2 = x\}}{36}, \quad x = 2, 3, \ldots, 12.$$

x	2	3	4	5	6	7	8	9	10	11	12
Ways	1	2	3	4	5	6	5	4	3	2	1

PMF of the Sum of Two Fair Dice



Cumulative Distribution Function

Definition

The cumulative distribution function (CDF) of X is

$$F_X(x) = \mathbb{P}(X \leq x).$$

For a discrete random variable

$$F_X(x) = \sum_{t \leq x} p_X(t).$$

Example

If X is the outcome of a fair die, then

$$F_X(3) = \mathbb{P}(X \leq 3) = \frac{3}{6} = \frac{1}{2}.$$

Expected Value

Definition

The **expected value** or **mean** of a discrete random variable X is

$$\mathbb{E}[X] = \sum_x x p_X(x).$$

- It is a weighted average of the possible values of X.
- Values with higher probability contribute more to the mean.

Example: Expected Value of a Fair Die

Let X be the outcome of a fair die.

$$\mathbb{E}[X] = \sum_{x=1}^{6} x\mathbb{P}(X = x).$$

Substitute the PMF:

$$\mathbb{E}[X] = 1 \cdot \frac{1}{6} + 2 \cdot \frac{1}{6} + 3 \cdot \frac{1}{6} + 4 \cdot \frac{1}{6} + 5 \cdot \frac{1}{6} + 6 \cdot \frac{1}{6}.$$

$$\mathbb{E}[X] = \frac{1+2+3+4+5+6}{6} = \frac{21}{6} = 3.5.$$

Example: Expected Heads in Three Tosses

For $Y =$ number of heads in three fair coin tosses:

y	$p_Y(y)$
0	$1/8$
1	$3/8$
2	$3/8$
3	$1/8$

$$\mathbb{E}[Y] = 0 \cdot \frac{1}{8} + 1 \cdot \frac{3}{8} + 2 \cdot \frac{3}{8} + 3 \cdot \frac{1}{8} = \frac{12}{8} = 1.5.$$

Expected Value Rule

If $Y = g(X)$, then

$$\mathbb{E}[g(X)] = \sum_x g(x) p_X(x).$$

Example

Let X be the outcome of a fair die. Find $\mathbb{E}[X^2]$.

$$\mathbb{E}[X^2] = \sum_{x=1}^{6} x^2 \frac{1}{6} = \frac{1^2 + 2^2 + 3^2 + 4^2 + 5^2 + 6^2}{6} = \frac{91}{6}.$$

Variance

Definition

The variance of X measures how spread out the values of X are around the mean.

$$\text{Var}(X) = \mathbb{E}[(X - \mu)^2], \quad \mu = \mathbb{E}[X].$$

Useful computational formula

$$\text{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2.$$

Standard deviation

$$\sigma_X = \sqrt{\text{Var}(X)}.$$

Example: Variance of a Fair Die

From the previous example,

$$\mathbb{E}[X] = \frac{7}{2}, \quad \mathbb{E}[X^2] = \frac{91}{6}.$$

Therefore,

$$\mathrm{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2.$$

Substitute:

$$\mathrm{Var}(X) = \frac{91}{6} - \left(\frac{7}{2}\right)^2 = \frac{91}{6} - \frac{49}{4} = \frac{35}{12}.$$

$$\sigma_X = \sqrt{\frac{35}{12}} \approx 1.708.$$

Properties of Expectation and Variance

Linearity of expectation

For constants a, b,

$$\mathbb{E}[aX + b] = a\mathbb{E}[X] + b.$$

For random variables X and Y,

$$\mathbb{E}[X + Y] = \mathbb{E}[X] + \mathbb{E}[Y].$$

Variance of a linear transformation

$$\mathrm{Var}(aX + b) = a^2 \,\mathrm{Var}(X).$$

Bernoulli Random Variable

Definition

A Bernoulli random variable models one trial with two outcomes: success or failure.

$$X = \begin{cases} 1, & \text{with probability } p, \\ 0, & \text{with probability } 1-p. \end{cases}$$

$$\mathbb{P}(X=1) = p, \quad \mathbb{P}(X=0) = 1-p.$$

$$\mathbb{E}[X] = p, \quad \mathrm{Var}(X) = p(1-p).$$

Example: Bernoulli Distribution

A system request succeeds with probability 0.9. Let

$$X = \begin{cases} 1, & \text{request succeeds,} \\ 0, & \text{request fails.} \end{cases}$$

Then $X \sim \text{Bernoulli}(0.9)$.

$$\mathbb{P}(X = 1) = 0.9, \quad \mathbb{P}(X = 0) = 0.1.$$

$$\mathbb{E}[X] = 0.9,$$

$$\text{Var}(X) = 0.9(1 - 0.9) = 0.09.$$

Discrete Uniform Random Variable

Definition

X is discrete uniform on $\{a, a+1, \ldots, b\}$ if all values are equally likely.

$$p_X(x) = \frac{1}{b-a+1}, \quad x = a, a+1, \ldots, b.$$

Fair die

A fair die is uniform on $\{1, 2, 3, 4, 5, 6\}$.

$$p_X(x) = \frac{1}{6}.$$

Binomial Random Variable

Definition

$X \sim \text{Binomial}(n, p)$ counts the number of successes in n independent Bernoulli trials, each with success probability p.

$$\mathbb{P}(X = k) = \binom{n}{k} p^k (1-p)^{n-k}, \quad k = 0, 1, \ldots, n.$$

$$\mathbb{E}[X] = np, \qquad \text{Var}(X) = np(1-p).$$

Use binomial when

There is a fixed number of trials, two outcomes per trial, independent trials, and the same success probability each time.

Binomial Example 1: True/False Quiz

A quiz has 5 true/false questions. A student guesses randomly. Let $X =$ number of correct answers.

$$X \sim \text{Binomial}\left(5, \frac{1}{2}\right).$$

Question

Find $\mathbb{P}(X = 3)$.

$$\mathbb{P}(X = 3) = \binom{5}{3}\left(\frac{1}{2}\right)^3\left(\frac{1}{2}\right)^2 = 10\left(\frac{1}{2}\right)^5 = \frac{10}{32} = 0.3125.$$

Binomial Example 2: Defective Items

A factory produces items. Each item is defective with probability 0.02. A sample of 10 items is selected. Let

$$X = \text{number of defective items.}$$

$$X \sim \text{Binomial}(10, 0.02).$$

Question

Find $\mathbb{P}(X \leq 1)$.

$$\mathbb{P}(X \leq 1) = \mathbb{P}(X = 0) + \mathbb{P}(X = 1).$$

$$= (0.98)^{10} + \binom{10}{1}(0.02)(0.98)^{9} \approx 0.9838.$$

Binomial Example 3: Successful Logins

A user login succeeds with probability 0.7. There are 8 independent login attempts. Let

$$X = \text{number of successful logins}, \qquad X \sim \text{Binomial}(8, 0.7).$$

Questions

Find $\mathbb{E}[X]$ and $\mathbb{P}(X \geq 6)$.

$$\mathbb{E}[X] = np = 8(0.7) = 5.6.$$

Since $X \geq 6$ means X can be 6, 7 or 8,

$$\begin{aligned}
\mathbb{P}(X \geq 6) &= \mathbb{P}(X = 6) + \mathbb{P}(X = 7) + \mathbb{P}(X = 8) \\
&= \binom{8}{6}(0.7)^6(0.3)^2 + \binom{8}{7}(0.7)^7(0.3)^1 + \binom{8}{8}(0.7)^8(0.3)^0 \\
&\approx 0.2965 + 0.1977 + 0.0576 = 0.5518.
\end{aligned}$$

Geometric Random Variable

Definition

$X \sim \text{Geometric}(p)$ counts the trial number on which the first success occurs.

$$\mathbb{P}(X = k) = (1-p)^{k-1}p, \quad k = 1, 2, 3, \ldots$$

$$\mathbb{E}[X] = \frac{1}{p}.$$

Use geometric when

We repeat independent trials until the first success occurs.

Geometric Distribution and Geometric Progression

For $X \sim \text{Geometric}(p)$, the probabilities are

$$\mathbb{P}(X = 1) = p,$$

$$\mathbb{P}(X = 2) = (1 - p)p,$$

$$\mathbb{P}(X = 3) = (1 - p)^2 p,$$

$$\mathbb{P}(X = 4) = (1 - p)^3 p, \ldots$$

Key idea

These probabilities form a **geometric progression** with common ratio $1 - p$.

Geometric Example 1: Network Packet

A network packet is successfully transmitted with probability 0.8 on each attempt. Let $X =$ attempt number on which the first success occurs.

$$X \sim \text{Geometric}(0.8).$$

Questions

Find $\mathbb{P}(X = 3)$ and $\mathbb{E}[X]$.

$$\mathbb{P}(X = 3) = (1 - 0.8)^{3-1}(0.8) = 0.2^2 \times 0.8 = 0.032.$$

$$\mathbb{E}[X] = \frac{1}{0.8} = 1.25.$$

Geometric Example 2: First Answered Call

A call is answered with probability 0.25 each time it is made. Let

$$X = \text{number of calls until the first answered call.}$$

$$X \sim \text{Geometric}(0.25).$$

Questions

Find $\mathbb{P}(X = 4)$ and $\mathbb{E}[X]$.

$$\mathbb{P}(X = 4) = (1 - 0.25)^{4-1}(0.25) = 0.75^3(0.25) \approx 0.1055.$$

$$\mathbb{E}[X] = \frac{1}{0.25} = 4.$$

Geometric Example 3: First Correct Password

A password attempt is correct with probability 0.3. Let

$$X = \text{attempt number on which the first correct password occurs.}$$

$$X \sim \text{Geometric}(0.3).$$

Question

Find the probability that the first success occurs within the first 3 attempts.

$$\mathbb{P}(X \leq 3) = 1 - \mathbb{P}(\text{first 3 attempts fail}).$$

$$\mathbb{P}(X \leq 3) = 1 - (0.7)^3 = 0.657.$$

Poisson Random Variable

Simplified description

A **Poisson random variable** counts how many times an event happens in a fixed interval of time, space or area.

Typical examples

- Number of calls received in one hour.
- Number of emails received in one minute.
- Number of defects on one page of printed text.
- Number of requests arriving at a server in one minute.

Key idea

Use Poisson when events occur independently and at an average rate.

Poisson Distribution: Formula and Mean

Definition

If $X \sim \text{Poisson}(\lambda)$, then

$$\mathbb{P}(X = k) = \frac{e^{-\lambda}\lambda^k}{k!}, \quad k = 0, 1, 2, \ldots$$

- λ is the average number of events in the interval.
- k is the number of events we want to calculate.

Mean and variance

$$\mathbb{E}[X] = \lambda, \qquad \text{Var}(X) = \lambda.$$

Poisson Mean as np

Connection with binomial

If a binomial model has large n and small p, then

$$X \sim \text{Binomial}(n, p) \approx \text{Poisson}(\lambda),$$

where

$$\lambda = np.$$

Important

In this approximation, the Poisson expected value can be calculated as

$$\mathbb{E}[X] = \lambda = np.$$

Example

If $n = 200$ and $p = 0.006$, then $\lambda = np = 1.2$.

Poisson Example 1: Server Requests

A server receives an average of 3 requests per minute. Let

$$X = \text{number of requests in one minute.}$$

$$X \sim \text{Poisson}(3).$$

Question

Find $\mathbb{P}(X = 2)$.

$$\mathbb{P}(X = 2) = \frac{e^{-3}3^2}{2!} = \frac{e^{-3} \times 9}{2} \approx 0.224.$$

Interpretation

There is about a 22.4% chance of receiving exactly two requests in one minute.

Poisson Example 2: Errors on a Page

A printed page has an average of 0.4 errors per page. Let

$$X = \text{number of errors on one page.}$$

$$X \sim \text{Poisson}(0.4).$$

Questions

Find $\mathbb{P}(X = 0)$ and $\mathbb{P}(X \leq 1)$.

$$\mathbb{P}(X = 0) = \frac{e^{-0.4}(0.4)^0}{0!} = e^{-0.4} \approx 0.6703.$$

$$\mathbb{P}(X \leq 1) = \mathbb{P}(X = 0) + \mathbb{P}(X = 1) = e^{-0.4} + 0.4e^{-0.4} \approx 0.9384.$$

Poisson Example 3: Calls per Hour

A help desk receives an average of 4 calls per hour. Let

$$X = \text{number of calls in one hour.}$$

$$X \sim \text{Poisson}(4).$$

Question

Find $\mathbb{P}(X \geq 6)$.

$$\mathbb{P}(X \geq 6) = 1 - \mathbb{P}(X \leq 5).$$

$$\mathbb{P}(X \geq 6) = 1 - \sum_{k=0}^{5} \frac{e^{-4} 4^k}{k!} \approx 0.2149.$$

Poisson Example 4: Binomial Approximation

A box has 200 nails. The probability that a nail is substandard is 0.006. Let

$$X = \text{number of substandard nails in the box.}$$

Approximation

$$X \sim \text{Binomial}(200, 0.006) \approx \text{Poisson}(\lambda),$$

$$\lambda = np = 200(0.006) = 1.2.$$

$$\mathbb{E}[X] = \lambda = np = 1.2.$$

Poisson Example 4: Probability Calculation

Using the Poisson approximation from the previous slide,

$$X \sim \text{Poisson}(1.2).$$

Question

Find the probability that the box contains at most two substandard nails.

$$\mathbb{P}(X \leq 2) = \mathbb{P}(X = 0) + \mathbb{P}(X = 1) + \mathbb{P}(X = 2).$$

$$\mathbb{P}(X \leq 2) = e^{-1.2} + 1.2e^{-1.2} + \frac{1.2^2 e^{-1.2}}{2!} = 2.92e^{-1.2} \approx 0.8795.$$

Two Random Variables

Sometimes one experiment produces more than one number.

Example: toss two fair coins

Let

$$X = \text{number of heads}, \qquad Y = \begin{cases} 1, & \text{first toss is H}, \\ 0, & \text{first toss is T}. \end{cases}$$

Outcome	HH	HT	TH	TT
X	2	1	1	0
Y	1	1	0	0

Key idea

X and Y are different random variables, but they come from the same experiment.

Joint Probability Mass Function

Definition

The **joint PMF** gives the probability that two random variables take values together:

$$p_{X,Y}(x, y) = \mathbb{P}(X = x \text{ and } Y = y).$$

Simple meaning

$p_{X,Y}(1, 0)$ means:

$$\mathbb{P}(X = 1 \text{ and } Y = 0).$$

It is not about X alone or Y alone. It is about the **pair** (X, Y).

Table idea

A joint PMF is usually easiest to read as a table. Each cell is one joint probability.

Joint PMF Table: Two Coin Tosses

For the two-coin example, each outcome has probability $1/4$.

	$X=0$	$X=1$	$X=2$	Row total
$Y=0$	$1/4$	$1/4$	0	$1/2$
$Y=1$	0	$1/4$	$1/4$	$1/2$
Column total	$1/4$	$1/2$	$1/4$	1

How to read one cell

$$p_{X,Y}(1,0) = \mathbb{P}(X=1, Y=0) = \frac{1}{4}.$$

This corresponds to outcome TH: one head and the first toss is tail.

Marginal Probability Mass Function

Definition

A **marginal PMF** gives the probability distribution of one random variable only.

Simple meaning

From the joint PMF table, we can answer questions about X alone or Y alone.

Remember

Marginalizing means: **keep one variable and ignore the other by adding over it.**

Marginal PMF Formula

Marginal PMF of X

To get the distribution of X alone, add over all possible values of Y:

$$p_X(x) = \sum_y p_{X,Y}(x, y).$$

Marginal PMF of Y

To get the distribution of Y alone, add over all possible values of X:

$$p_Y(y) = \sum_x p_{X,Y}(x, y).$$

Table rule

Column totals give $p_X(x)$; row totals give $p_Y(y)$.

Finding the Marginal PMF of X

Use the column totals from the joint PMF table.

	$X=0$	$X=1$	$X=2$
$Y=0$	$1/4$	$1/4$	0
$Y=1$	0	$1/4$	$1/4$
$p_X(x)$	$1/4$	$1/2$	$1/4$

$$p_X(1) = p_{X,Y}(1,0) + p_{X,Y}(1,1) = \frac{1}{4} + \frac{1}{4} = \frac{1}{2}.$$

Interpretation

There is probability $1/2$ of getting exactly one head in two tosses.

Finding the Marginal PMF of Y

Use the row totals from the joint PMF table.

	$X=0$	$X=1$	$X=2$	$p_Y(y)$
$Y=0$	$1/4$	$1/4$	0	$1/2$
$Y=1$	0	$1/4$	$1/4$	$1/2$

$$p_Y(0) = p_{X,Y}(0,0) + p_{X,Y}(1,0) + p_{X,Y}(2,0) = \frac{1}{4} + \frac{1}{4} + 0 = \frac{1}{2}.$$

Interpretation

There is probability $1/2$ that the first toss is tail.

Joint PMF vs Marginal PMF

Joint PMF	**Marginal PMF**
Uses two variables: X and Y	Uses one variable only: X or Y
Answers questions like $\mathbb{P}(X = x, Y = y)$	Answers questions like $\mathbb{P}(X = x)$ or $\mathbb{P}(Y = y)$
Each table cell is a joint probability	Row or column totals are marginal probabilities

Remember

Joint = together. Marginal = total after ignoring one variable.

Conditional PMF: Simple Idea

Meaning

A **conditional PMF** is a probability distribution after we are given extra information.

Think of a joint table

If we are told $Y = y$, then:

- keep only the row/column where $Y = y$;
- ignore the other values of Y;
- divide by the total probability of $Y = y$.

In words

$$\mathbb{P}(X = x \mid Y = y) = \frac{\text{probability of both } X = x \text{ and } Y = y}{\text{probability of } Y = y}.$$

Conditional PMF: Formula

Definition

If $p_Y(y) > 0$, then

$$p_{X|Y}(x \mid y) = \mathbb{P}(X = x \mid Y = y) = \frac{p_{X,Y}(x, y)}{p_Y(y)}.$$

Rearranged form

$$p_{X,Y}(x, y) = p_Y(y) p_{X|Y}(x \mid y).$$

Key phrase

Conditional PMF means: **use the joint probability, then divide by the marginal probability.**

Example: Conditional PMF from the Joint Table

Recall:

$$X = \text{number of heads in two tosses,} \qquad Y = \begin{cases} 1, & \text{first toss is H,} \\ 0, & \text{first toss is T.} \end{cases}$$

	$X=0$	$X=1$	$X=2$	$p_Y(y)$
$Y=0$	$1/4$	$1/4$	0	$1/2$
$Y=1$	0	$1/4$	$1/4$	$1/2$

Question

Given that the first toss is tail, find $\mathbb{P}(X = 1 \mid Y = 0)$.

Solution: Conditional PMF Example

We want

$$\mathbb{P}(X = 1 \mid Y = 0).$$

Step 1: identify the needed probabilities

$$p_{X,Y}(1,0) = \frac{1}{4}, \qquad p_Y(0) = \frac{1}{2}.$$

Step 2: apply the formula

$$p_{X|Y}(1 \mid 0) = \frac{p_{X,Y}(1,0)}{p_Y(0)} = \frac{\frac{1}{4}}{\frac{1}{2}} = \frac{1}{2}.$$

Meaning

If we already know the first toss was tail, then there is probability $1/2$ of getting exactly one head.

Total Probability for PMFs

Formula

$$p_X(x) = \sum_y p_Y(y) p_{X|Y}(x \mid y).$$

Simple meaning

To find the probability of $X = x$, split the problem into cases for Y:

$$\begin{aligned}\mathbb{P}(X = x) = \mathbb{P}(X = x \mid Y = 0)\mathbb{P}(Y = 0) \\ + \mathbb{P}(X = x \mid Y = 1)\mathbb{P}(Y = 1) + \cdots\end{aligned}$$

Short memory aid

Total probability = **case 1 contribution + case 2 contribution + ...**

Example: Total Probability from Conditional PMFs

Use the same two-coin example. Find $p_X(1)$.

From the conditional distributions

$$p_{X|Y}(1 \mid 0) = \frac{1}{2}, \qquad p_{X|Y}(1 \mid 1) = \frac{1}{2},$$

$$p_Y(0) = \frac{1}{2}, \qquad p_Y(1) = \frac{1}{2}.$$

Apply total probability

$$p_X(1) = p_Y(0)p_{X|Y}(1 \mid 0) + p_Y(1)p_{X|Y}(1 \mid 1).$$

$$p_X(1) = \left(\frac{1}{2}\right)\left(\frac{1}{2}\right) + \left(\frac{1}{2}\right)\left(\frac{1}{2}\right) = \frac{1}{2}.$$

Independence of Discrete Random Variables

Definition

Two discrete random variables X and Y are **independent** if

$$p_{X,Y}(x, y) = p_X(x)p_Y(y) \quad \text{for all } x, y.$$

Simple meaning

Knowing the value of Y does **not** change the probabilities for X.

$$p_{X|Y}(x \mid y) = p_X(x).$$

Memory aid

Independent means: **together probability = first probability $\times$ second probability.**

Example: Independent Random Variables

Toss two fair coins. Define

$$A = \begin{cases} 1, & \text{first toss is H,} \\ 0, & \text{first toss is T,} \end{cases} \qquad B = \begin{cases} 1, & \text{second toss is H,} \\ 0, & \text{second toss is T.} \end{cases}$$

	$A=0$	$A=1$	$p_B(b)$
$B=0$	$1/4$	$1/4$	$1/2$
$B=1$	$1/4$	$1/4$	$1/2$
$p_A(a)$	$1/2$	$1/2$	1

Solution: Independence Example

Check one cell:

$$p_{A,B}(1,1) = \frac{1}{4}.$$

The marginal probabilities are

$$p_A(1) = \frac{1}{2}, \qquad p_B(1) = \frac{1}{2}.$$

Therefore,

$$p_A(1)p_B(1) = \left(\frac{1}{2}\right)\left(\frac{1}{2}\right) = \frac{1}{4}.$$

Conclusion

For every cell, $p_{A,B}(a,b) = p_A(a)p_B(b)$. So A and B are independent.

Conditional Probability Under Independence

If A and B are independent, then knowing B does not change A.

Formula

$$p_{A|B}(a \mid b) = p_A(a).$$

Using the coin example

$$\mathbb{P}(A = 1 \mid B = 1) = \frac{p_{A,B}(1,1)}{p_B(1)} = \frac{\frac{1}{4}}{\frac{1}{2}} = \frac{1}{2}.$$

But

$$\mathbb{P}(A = 1) = \frac{1}{2}.$$

Meaning

Knowing the second toss is head does not change the probability that the first toss is head.

Example: Not Independent

Return to

$$X = \text{number of heads}, \qquad Y = \text{whether the first toss is H.}$$

From the joint table,

$$p_{X,Y}(2,1) = \frac{1}{4}.$$

But the marginal probabilities are

$$p_X(2) = \frac{1}{4}, \qquad p_Y(1) = \frac{1}{2}.$$

So

$$p_X(2)p_Y(1) = \left(\frac{1}{4}\right)\left(\frac{1}{2}\right) = \frac{1}{8} \neq \frac{1}{4}.$$

Conclusion

X and Y are **not independent**. Knowing that the first toss is head affects the number of heads.